VAST 2022 - Challenge 3: Economic

INTRODUCTION

Engagement, Ohio is going through some big changes, going from a quiet town to a popular spot. To prep for this growth, the city is getting the residents involved in urban planning. About 1000 folks are using a city app to share info about where they go, what they spend, and what they buy. The city planners wants to use this data to spruce up the place, especially since they snagged a fat renewal grant. We, as a visual analytics expert, are part of the team trying to make sense of all this resident-provided data. We aim to give the city some practical insights to help them make smart choices for their community makeover.

SYSTEM DESIGN

We've used different types of charts to offer a complete understanding of data on restaurants, pubs, residents, and employers in the city. Each graph has its own set of marks, channels, and interactive features, making it easy for city planners to dive into the data and make smart decisions. It's all about getting quick, accurate insights that help steer the ship in the right direction.

DATA SET

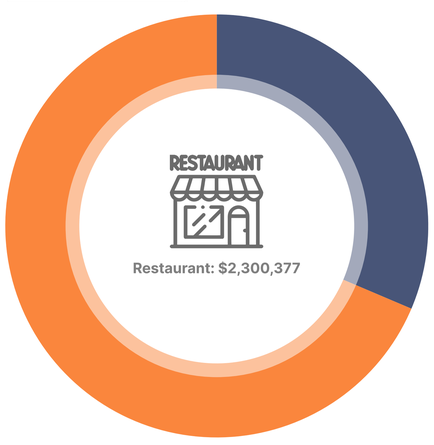
We've got a rich dataset, capturing detailed logs of over a thousand people's activities every 5 minutes for 15 months. It covers everything from buildings, employers, jobs, residents, to specifics from pubs, restaurants, check-ins, financial transactions, and travel journals. We preprocessed and wrangled the data, creating task-specific CSV files. To spice it up, we calculated extra details like Turnover, revenue, monthly income, expenses, and savings using Python and some math magic. And for flexibility in analyses, we added a dash of browser-based preprocessing. Ready for a deep dive into the city's story!

Q1

Over the period covered by the dataset, which businesses appear to be more prosperous? Which appear to be struggling?

View By

Revenue



Total Revenue by Businesses

Overview

This donut chart breaks down two important things to check how well businesses in the city are doing. First up is the money side, showing the cash flow for places like pubs and restaurants. If you're not into the money talk, you can switch it up and look at the foot traffic instead – whatever floats your boat.

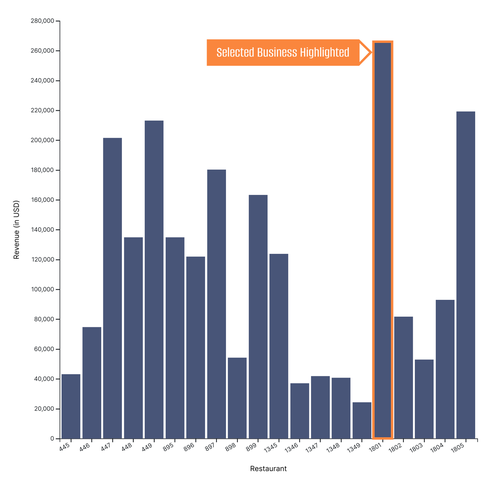
Design

We've used bright colors to tell apart different types of businesses. This color code stays consistent in the other charts, so you won't get lost. We've kept the important stuff, like how much money places make and how many people go there, in easy-to-read formats.

Click on any section, and the next chart spills the beans with more details about that specific kind of business. No secrets here!

Interaction

The chart is easy to play around with. You can switch between money and foot traffic views with a click. It's all about making it work for you. Hover over any section, and you'll see a little icon representing the business type and how much money they pulled in during the selected time.



Revenue generated by Individual Restaurants

Overview

This bar chart provides insights into the average metric (revenue/footfall) specific to the chosen business category. It breaks down individual businesses, offering a detailed view of each one's performance relative to others over the selected timeline.

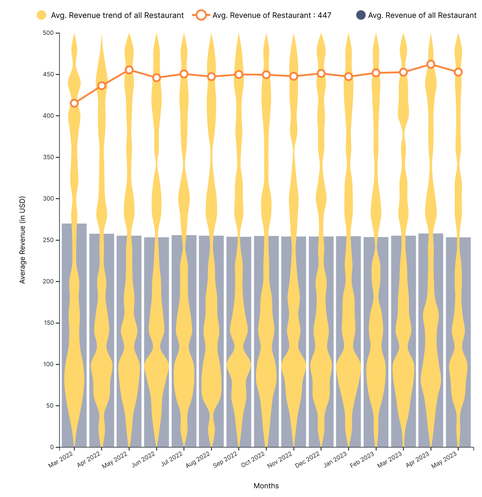
Design

To keep things clear, we've used consistent colors to represent similar businesses. Click on a bar for a specific business, and it stands out with a different color stroke, and you'll be taken to the next chart that provides more details for this specific business.

Upon clicking a business's bar, the subsequent chart unfolds, providing a deeper dive into the details of that particular business.

Interaction

Hover over a bar, and a tooltip appears, sharing the exact metric values for that business. Click on a bar, and it leads you to a more detailed chart specific to that business.



Month-wise Revenue generated by Restaurant

Overview

This chart offers a comprehensive view of a business's prosperity, loaded with key metrics for an overarching comparison.

- The orange line chart illustrates the metric (revenue/footfall) for the chosen business from the preceding chart.
- The blue bar chart details the average metric for the selected business category on a monthly basis within the chosen timeline, helping you gauge your business against competitors.
- The yellow violin chart, a unique feature, depicts the distribution trend of the metric across all competitors, allowing you to understand the overall trend and your business's performance in that context.

Design

To maintain clarity, we've stuck to the same colors for similar businesses. Carefully chosen colors distinguish vital metrics for enhanced comprehension.

Interaction

Hover over any element—a line, bar, or violin—and a tooltip appears, offering precise metric values for that specific business. This interactive feature enhances user engagement and understanding

Innovative Design

Q2

How does the financial health of the residents change over the period covered by the dataset? How do wages compare to the overall cost of living in Engagement? Are there groups that appear to exhibit similar patterns?

Overview

This stream graph gives a quick look at how the city's residents are doing financially over time. It shows trends in their financial status changes providing user with a quick overview of resident's financial health.

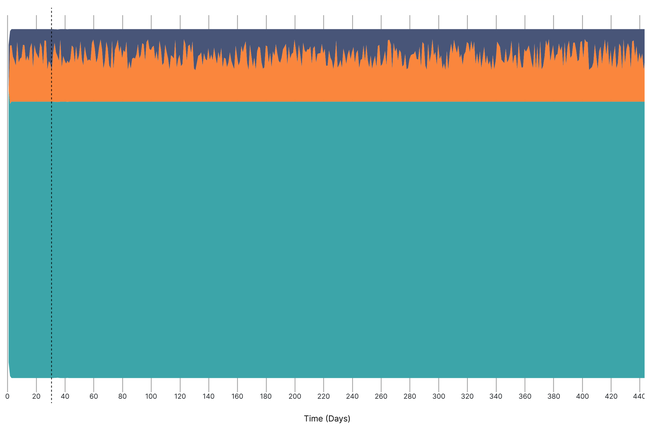
Design

We've used colors that make sense culturally, so it's easy for users to get it. The color legend above the graph helps users understand the chart. Plus, we've set the axis to days for a more detailed view.

Interaction

We added a tooltip popover for users to see exact counts for different financial status categories. Also, a vertical line helps users know exactly which days they're looking at.

Stable Unstable Unknown

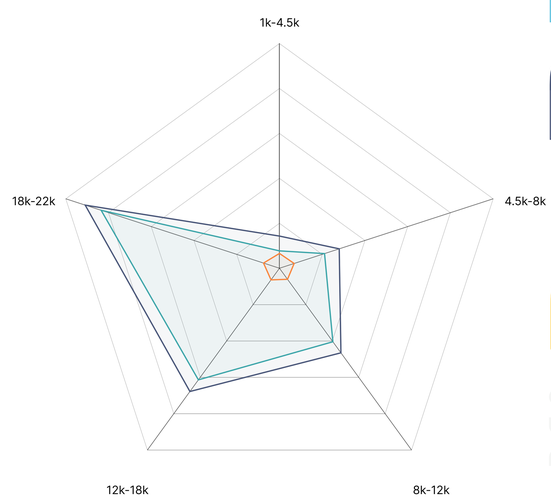


Financial Health of residents

Group By

Education Level

Income
Expenses
Savings



Financial Habits of residents

Overview

This spider chart shows the average income, expenses, and savings (derived attribute) for residents. There are two grouping options that make the most logical sense, so users can get different insights into residents' financial habits.

Design

Users can switch on/off attributes displayed on the chart, customizing their experience based on what they need.

Upon changing the Group By attribute subsequent chart's grouping updates, providing a deeper dive into the details of that particular group.

Interaction

The chart has an informative tooltip showing exact values for comparison within financial categories.

Overview

This line chart provides intricate details on residents' spending habits, color-coded by the groupings from the spider chart, throughout the selected timeline. It helps users spot patterns in how residents handle their money.

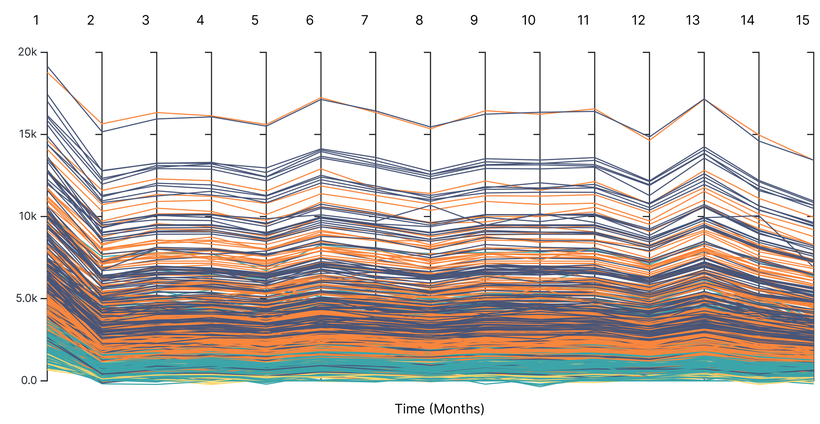
Graduate
Bachelors
HighSchool
Low

Design

The chart breaks down savings for each resident within the user chosen timeline grouped by selected metric.

Interaction

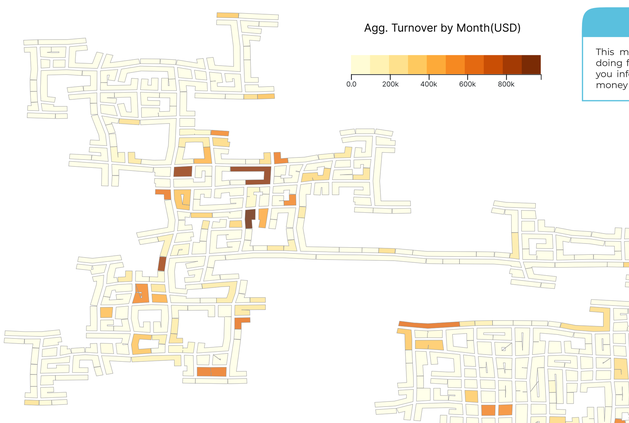
Hovering over a line shows an informative tooltip with exact values. Also, the chart highlights residents with similar education backgrounds or salary ranges, depending on the chosen metric.



Monthly Savings of residents

Q3

Describe the health of the various employers within the city limits. What employment patterns do you observe?



Overview

This map lays out how different employers in the city are doing financially, using colors to show their turnover. It gives you info on where each employer is located and how much money they're making within a specific time frame.

Design

The chart gives a quick look at where employers are in the city and how they're doing financially during the chosen timeline.

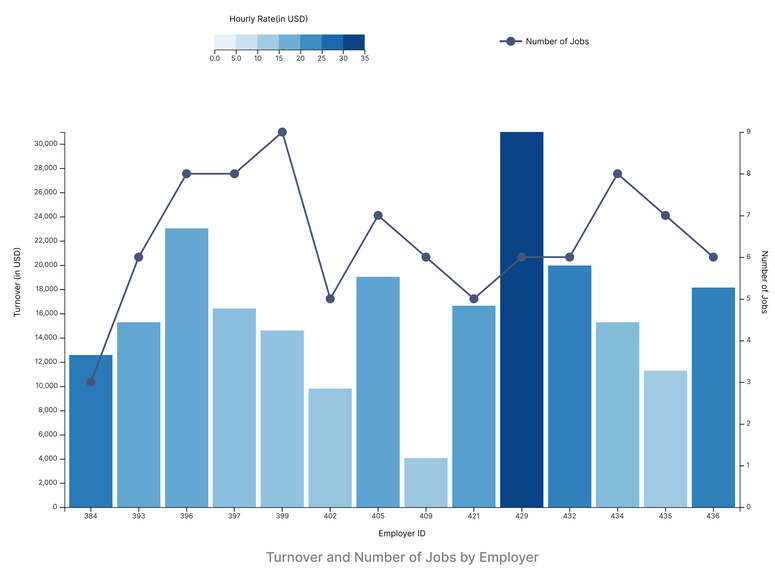
Interaction

The map offers a cool user experience by connecting to the next chart. You can brush over areas on the map to pick a region, which gets highlighted, and then the next chart pops up.

The next bar chart dives into the details of the employers in the selected region

Turnover By Employers

Interactive Chart with Region Brushing



Turnover and Number of Jobs by Employer

Overview

This line chart is a data-packed tool, focusing solely on employers in the region you picked on the map. The bars show how much money each employer in that region is making, with colors indicating their hourly rates. Plus, the line chart tells you how many jobs each employer usually posts, giving you some useful insights into the city's job scene.

Design

The design uses two axes to make it easy for user to get useful information.

Interaction

Clicking on any employer highlights it on the map chart, and a tooltip shows up with more information about that employer.

MEET THE TEAM

That made this awesome project

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